IGR in MFC: Integration and Preliminary MHD Results

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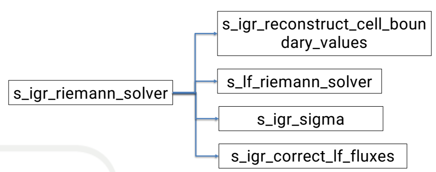
Project Overview

This OSPO VSIP project aims to integrate information geometric regularization (IGR) in the MFC codebase to use shared simulation infrastructure/subroutines with the goal of eliminating unnecessary hurdles to researchers interested in advancing IGR-related work.

Toward the end of the internship, we also tested a new IGR-MHD feature that utilizes the newly refactored IGR path.

Goals and Milestones

- **Milestones 1 & 2: IGR Integration**
  The two milestones are defined as follows:
  milestone 1: inviscid IGR refactor
  milestone 2: viscous IGR refactor
  The final refactored path follows the diagram:

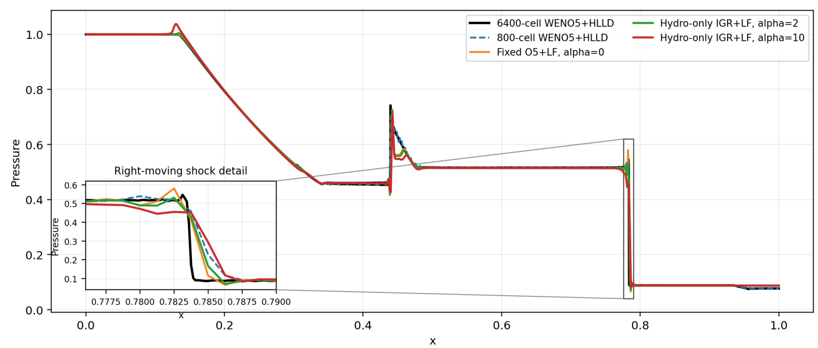


  However, due to the nonlinearity of the reconstruction order:
  Recon(Prim(Cons)) != Prim(Recon(Cons))
  (normal path order)   (original IGR order)
  Any changes beyond the *igr-integration* branch will not preserve numerical equivalence to roundoff. As a result, we moved further integration to the *igr-mhd* experimental branch.

- **Final Goal: Preliminary IGR–MHD Capability**
  We establish some preliminary results demonstrating the feasibility of using IGR to regularize MHD equations.

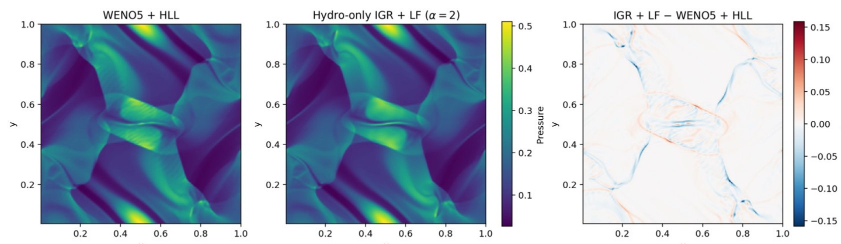
Highlights and Accomplishments

Brio--Wu pressure at t=0.2: regularization of polynomial overshoot



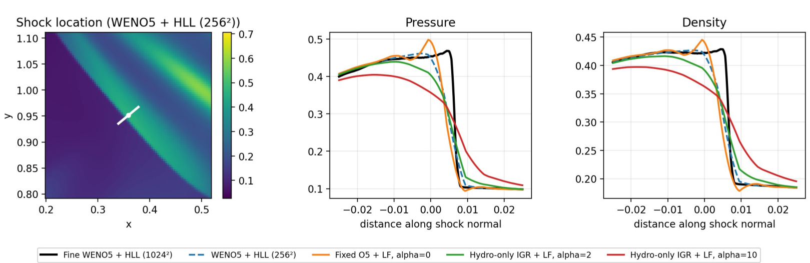
Brio−Wu demonstrates IGR's intended regularization mechanism. With the same fifth-order fixed reconstruction, positive alpha damps the pressure overshoot near the right-moving shock.

Orszag--Tang thermal pressure at t=0.5 (256²)



For Orszag-Tang, IGR damps the fixed-polynomial overshoot at the selected front and delays the first negative reconstructed face from t=0.2 to t=0.77. It also reduces RHS runtime by about 20% relative to WENO5+HLL, at the cost of additional smearing.

Orszag--Tang shock profile at t=0.2



Open-Source Outcomes

- **IGR Integration**
  While integrating IGR provides easier access to the method, it also exposes a tradeoff between fused subroutines and more modular components. The refactored path runs roughly ~1.2x as fast as the original fused path and costs ~3x as much memory, although both in line with expectations.

- **IGR + MHD**
  The MHD experiments expose the expected trade-off: hydro-only IGR damps fixed-polynomial shock oscillations and reduces reconstruction cost but introduces additional smearing and does not consistently improve global accuracy.

Future Work

- **Resolve IGR Reconstruction Ordering**
  As noted earlier, in the experimental IGR-MHD branch, we changed the IGR reconstruction order to conservative -> primitive -> reconstruction which alters numerical behavior. We need to work with the team to figure out whether this is an acceptable change.

- **Derive Maxwell-Consistent IGR Formulation**
  The current MHD-IGR cases use existing hydro-only IGR formulation, ignoring Maxwell's IGR contribution. Although it seems to work empirically, we still need a complete derivation.